

Research Idea Generation Workflow Library

Expert Workflow Library

Ultimate AI Research Toolkit 2026 - Premium Edition
Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	Research prompts and ideation systems
[Review]	Literature synthesis and gap detection
[Write]	Paper writing and revision workflows
[Plan]	Publication and productivity operations

Table of Contents

1. Introduction
2. Instructions
3. Workflow Library
4. Summary

Introduction

This Research Idea Generation collection provides 120 expert research workflows engineered for high-rigor academic and technical delivery. Each workflow includes practical fields to help users move from strategy to publication-ready output.

Instructions

- Select workflows by category and difficulty to match project maturity.
- Replace variables in each workflow with your discipline, data, and resource constraints.
- Use Expected Output as the acceptance criteria for quality control.
- Store outputs in versioned folders to preserve decision history.

Workflow Library

1. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - practical deployment framing

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=benchmarking specialist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

2. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - method comparison clarity

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=theory-oriented reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Analysis

3. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - editorial risk reduction

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=methodology purist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Writing

4. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - venue-specific optimization

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=interdisciplinary committee member. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Publication

5. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - decision traceability

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=ethics compliance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

6. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - citation-strength positioning

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=reproducibility auditor. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Execution

7. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - defense-ready

reasoning

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=applied domain expert. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Compliance

8. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - high-confidence statistical reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=systems performance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

9. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - evidence-first writing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=industry impact reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Analysis

10. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - hypothesis stress testing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=editor focused on clarity. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Writing

11. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - results-to-insight translation

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=skeptical statistician. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Publication

12. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - major-revision readiness

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=benchmarking specialist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Methodology

13. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - replication-grade documentation

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=theory-oriented reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Execution

14. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - reviewer-resilient argumentation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=methodology purist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Compliance

15. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - ethics-by-design reporting

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain

data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=interdisciplinary committee member. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Reasoning

16. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - limitations-first transparency

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=ethics compliance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Rebuttal

17. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - cross-domain readability

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred

method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=reproducibility auditor. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Validation

18. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - impact-focused communication

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=applied domain expert. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Publication

19. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - publishable novelty framing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline

outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=systems performance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Methodology

20. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - practical deployment framing

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=industry impact reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Execution

21. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - method comparison clarity

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=editor focused on clarity. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Compliance

22. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - editorial risk reduction

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=skeptical statistician. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Reasoning

23. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - venue-specific optimization

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=benchmarking specialist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Rebuttal

24. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - decision traceability

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=theory-oriented reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Validation

25. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - citation-strength positioning

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=methodology purist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Optimization

26. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - defense-ready reasoning

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=interdisciplinary committee member. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Planning

27. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - high-confidence statistical reporting

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=ethics compliance reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and

milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Execution

28. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - evidence-first writing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=reproducibility auditor. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Compliance

29. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - hypothesis stress testing

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=applied domain expert. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Reasoning

30. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - results-to-insight translation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=systems performance reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Rebuttal

31. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - major-revision readiness

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=industry impact reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Validation

32. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - replication-grade documentation

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=editor focused on clarity. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Optimization

33. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - reviewer-resilient argumentation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=skeptical statistician. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Planning

34. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - ethics-by-design reporting

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=benchmarking specialist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Strategy

35. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - limitations-first transparency

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=theory-oriented reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Analysis

36. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - cross-domain readability

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=methodology purist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

37. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - impact-focused communication

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=interdisciplinary committee member. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Rebuttal

38. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - publishable novelty framing

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=ethics compliance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Validation

39. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - practical deployment framing

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=reproducibility auditor. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

40. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - method comparison clarity

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=applied domain expert. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Planning

41. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - editorial risk reduction

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=systems performance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

42. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - venue-specific optimization

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=industry impact reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Analysis

43. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - decision traceability

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=editor focused on clarity. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Writing

44. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - citation-strength positioning

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=skeptical statistician. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Publication

45. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - defense-ready reasoning

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=benchmarking specialist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Validation

46. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - high-confidence statistical reporting

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=digital health; dataset=public

genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=theory-oriented reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Optimization

47. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - evidence-first writing

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=methodology purist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Planning

48. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - hypothesis stress testing

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=interdisciplinary committee member. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Strategy

49. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - results-to-insight translation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=ethics compliance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Analysis

50. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - major-revision readiness

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=reproducibility auditor. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Writing

51. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - replication-grade documentation

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=applied domain expert. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Publication

52. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - reviewer-resilient argumentation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=systems performance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Methodology

53. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - ethics-by-design reporting

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=industry impact reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Execution

54. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - limitations-first transparency

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=editor focused on clarity. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Planning

55. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - cross-domain readability

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=skeptical statistician. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Strategy

56. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - impact-focused communication

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=benchmarking specialist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Analysis

57. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - publishable novelty framing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=theory-oriented reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Writing

58. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - practical deployment framing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=methodology purist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Publication

59. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - method comparison clarity

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=interdisciplinary committee member. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Methodology

60. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - editorial risk reduction

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=ethics compliance reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Execution

61. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - venue-specific optimization

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=reproducibility auditor. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Compliance

62. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - decision traceability

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=applied domain expert. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Reasoning

63. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - citation-strength

positioning

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=systems performance reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Analysis

64. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - defense-ready reasoning

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=industry impact reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Writing

65. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - high-confidence statistical reporting

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=editor focused on clarity. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Publication

66. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - evidence-first writing

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=skeptical statistician. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Methodology

67. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - hypothesis stress testing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=benchmarking specialist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Execution

68. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - results-to-insight translation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=theory-oriented reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Compliance

69. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - major-revision readiness

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist;

output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=methodology purist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Reasoning

70. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - replication-grade documentation

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=interdisciplinary committee member. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Rebuttal

71. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - reviewer-resilient argumentation

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget

reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=ethics compliance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Validation

72. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - ethics-by-design reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=reproducibility auditor. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Publication

73. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - limitations-first transparency

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=applied domain expert. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

74. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - cross-domain readability

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=systems performance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Execution

75. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - impact-focused communication

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=industry impact reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Compliance

76. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - publishable novelty framing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=editor focused on clarity. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

77. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - practical deployment framing - Variant 077

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=skeptical statistician. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Rebuttal

78. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - method comparison clarity - Variant 078

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=benchmarking specialist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Validation

79. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - editorial risk reduction - Variant 079

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=theory-oriented reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

80. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - venue-specific optimization - Variant 080

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=methodology purist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Planning

81. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - decision traceability - Variant 081

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=interdisciplinary committee member. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Execution

82. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - citation-strength positioning - Variant 082

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=ethics compliance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Compliance

83. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - defense-ready reasoning - Variant 083

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=reproducibility auditor. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Reasoning

84. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - high-confidence statistical reporting - Variant 084

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=applied domain expert. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Rebuttal

85. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - evidence-first writing - Variant 085

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=systems performance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Validation

86. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - hypothesis stress testing - Variant 086

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=industry impact reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Optimization

87. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - results-to-insight translation - Variant 087

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=editor focused on clarity. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Planning

88. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - major-revision readiness - Variant 088

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=skeptical statistician. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Strategy

89. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - replication-grade documentation - Variant 089

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=benchmarking specialist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Analysis

90. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - reviewer-resilient argumentation - Variant 090

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=theory-oriented reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Reasoning

91. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - ethics-by-design reporting - Variant 091

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=methodology purist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Rebuttal

92. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - limitations-first transparency - Variant 092

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=interdisciplinary committee member. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Validation

93. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - cross-domain readability - Variant 093

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=ethics compliance reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from an ethics compliance reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Optimization

94. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - impact-focused communication - Variant 094

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=reproducibility auditor. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Planning

95. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - publishable novelty framing - Variant 095

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=applied domain expert. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Strategy

96. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - practical deployment framing - Variant 096

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=systems performance reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Analysis

97. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - method comparison clarity - Variant 097

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=industry impact reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Writing

98. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - editorial risk reduction - Variant 098

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=editor focused on clarity. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Publication

99. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - venue-specific optimization - Variant 099

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=skeptical statistician. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Validation

100. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - decision traceability - Variant 100

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=benchmarking specialist. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Optimization

101. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - citation-strength positioning - Variant 101

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=theory-oriented reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Planning

102. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - defense-ready reasoning - Variant 102

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=methodology purist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Strategy

103. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - high-confidence statistical reporting - Variant 103

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=interdisciplinary committee member. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Analysis

104. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - evidence-first writing - Variant 104

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=ethics compliance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Writing

105. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - hypothesis stress testing - Variant 105

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Hypothesis Framing.

Prompt: Analyze a draft on Hypothesis Framing and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM Computing Surveys. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=reproducibility auditor. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Publication

106. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - results-to-insight translation - Variant 106

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Problem Selection.

Prompt: Interpret results for Problem Selection using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in Nature Communications. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=applied domain expert. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Methodology

107. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - major-revision readiness - Variant 107

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Feasibility Scan.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Feasibility Scan and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Expert Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=systems performance reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Execution

108. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - replication-grade documentation - Variant 108

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Novelty Discovery.

Prompt: Generate a publication-ready section for Novelty Discovery tuned for IEEE Transactions. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=industry impact reviewer. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Planning

109. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - reviewer-resilient argumentation - Variant 109

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Hypothesis Framing.

Prompt: Audit the experimental design for Hypothesis Framing. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM Computing Surveys reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=editor focused on clarity. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

110. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - ethics-by-design reporting - Variant 110

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Problem Selection.

Prompt: Create a LaTeX writing workflow for Problem Selection: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for Nature Communications submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=skeptical statistician. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Analysis

111. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - limitations-first transparency - Variant 111

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Feasibility Scan.

Prompt: Simulate a high-pressure Q&A; on Feasibility Scan. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=benchmarking specialist. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Writing

112. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - cross-domain readability - Variant 112

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Novelty Discovery.

Prompt: Act as a senior research editor. Build a contribution architecture for Novelty Discovery in early-stage thesis planning targeting IEEE Transactions. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=theory-oriented reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Publication

113. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - impact-focused communication - Variant 113

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Hypothesis Framing.

Prompt: Rewrite the methodology around Hypothesis Framing for first-author conference submission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM Computing Surveys. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=methodology purist. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

114. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - publishable novelty framing - Variant 114

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Problem Selection.

Prompt: Write a Nature-style abstract for a study on Problem Selection in interdisciplinary grant concept. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=interdisciplinary committee member. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Execution

115. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - practical deployment framing - Variant 115

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Feasibility Scan.

Prompt: Generate a figure strategy for Feasibility Scan with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Expert Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=ethics compliance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Compliance

116. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - method comparison clarity - Variant 116

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Novelty Discovery.

Prompt: Build an ethics checklist and mitigation plan for Novelty Discovery in early-stage thesis planning. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for IEEE Transactions review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=reproducibility auditor. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

117. Generate a Strong Research Gap: high-impact scope design for first-author conference submission (ACM Computing Surveys) - editorial risk reduction - Variant 117

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Hypothesis Framing.

Prompt: Design a publication strategy for Hypothesis Framing: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM Computing Surveys. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM Computing Surveys; reviewer persona=applied domain expert. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Framing.

Difficulty: Beginner

Category: Analysis

118. Design a Defensible Hypothesis: contribution stress test for interdisciplinary grant concept (Nature Communications) - venue-specific optimization - Variant 118

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Problem Selection.

Prompt: Assume Reviewer #2 challenges Problem Selection for Nature Communications. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Nature Communications; reviewer persona=systems performance reviewer. Example outcome: hypothesis matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready hypothesis matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Problem Selection.

Difficulty: Intermediate

Category: Writing

119. Map Breakthrough Research Opportunities: gap-to-hypothesis workflow for industry-sponsored research proposal (Elsevier Expert Systems) - decision traceability - Variant 119

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Feasibility Scan.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Feasibility Scan. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Expert Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Expert Systems; reviewer persona=industry impact reviewer. Example outcome: priority map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready priority map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feasibility Scan.

Difficulty: Advanced

Category: Publication

120. Create Novel Contributions: novelty pipeline for early-stage thesis planning (IEEE Transactions) - citation-strength positioning - Variant 120

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Novelty Discovery.

Prompt: Design a statistical analysis workflow for Novelty Discovery with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for IEEE Transactions. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Transactions; reviewer persona=editor focused on clarity. Example outcome: research roadmap with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready research roadmap that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Novelty Discovery.

Difficulty: Expert

Category: Methodology

Summary

Applying these 120 workflows systematically improves decision quality, writing clarity, and publication readiness across research cycles.